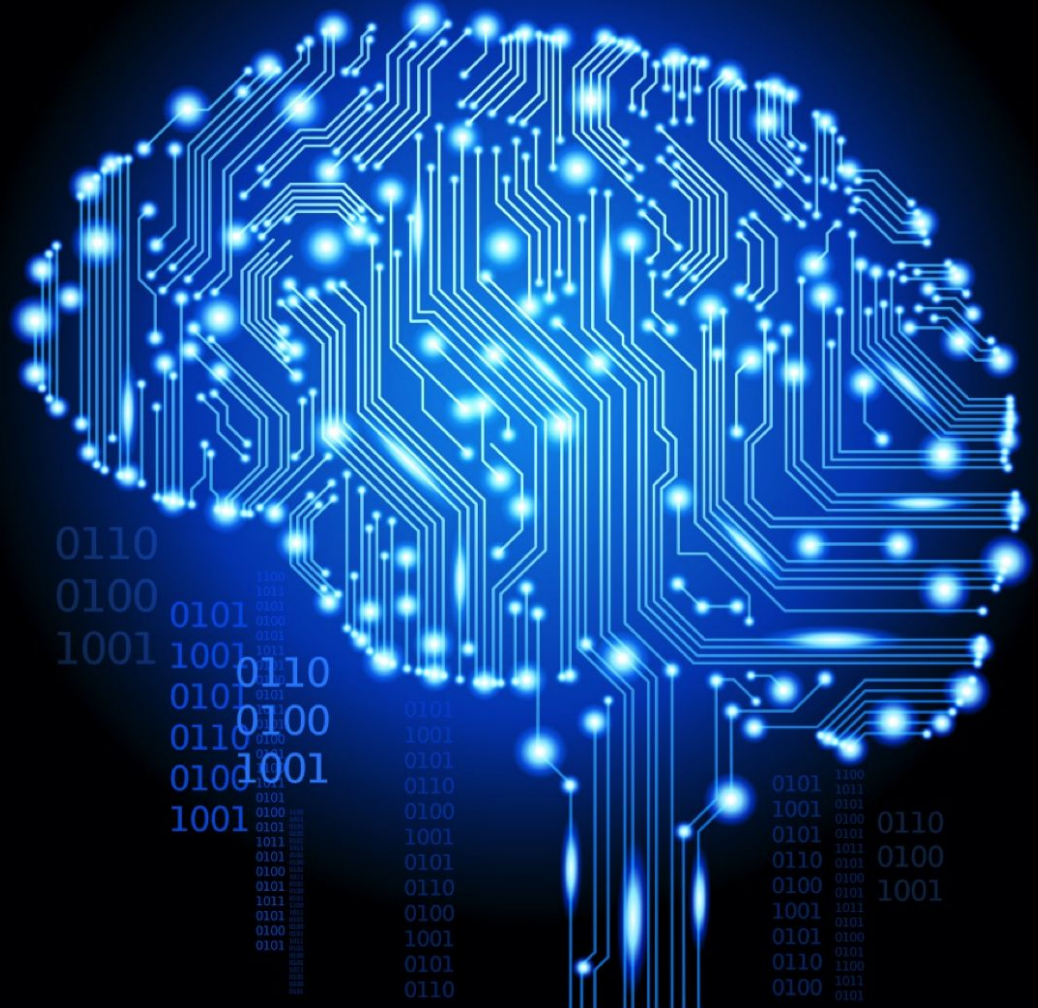


Introduction to DATA SCIENCE



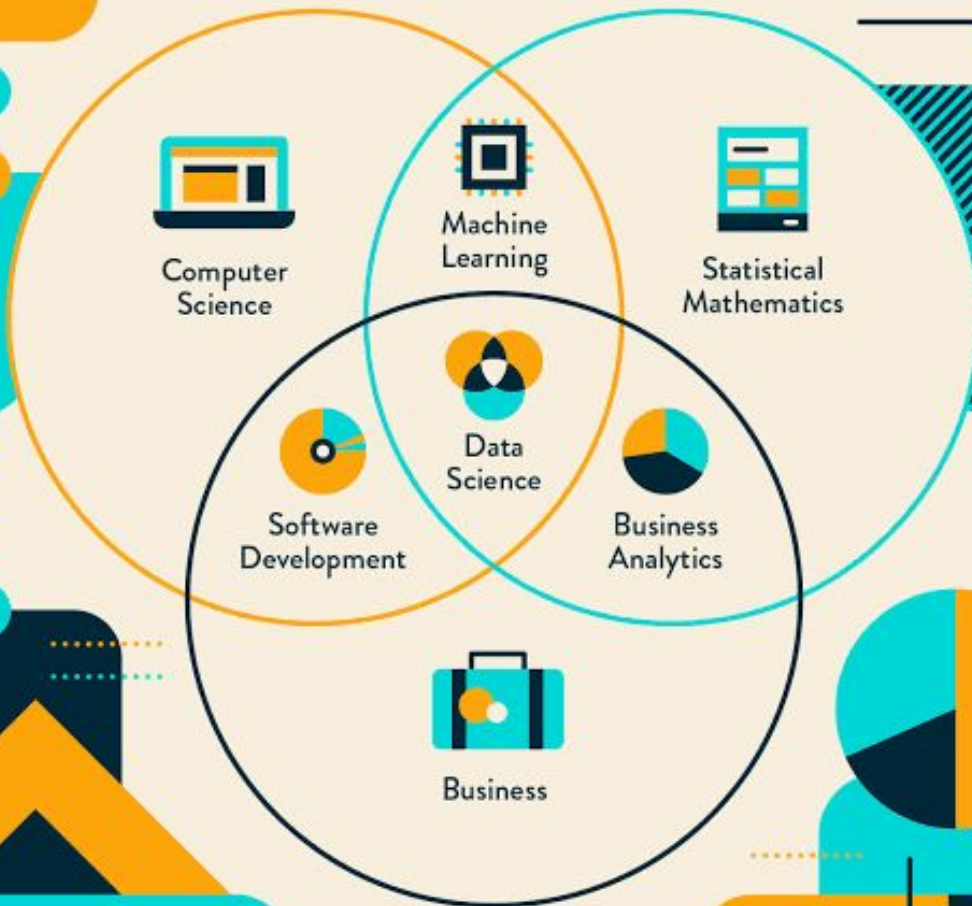
ANALYSIS VISUALIZATION SYSTEM

KNOWLEDGE METHODS

DATA SCIENCE

PROCESS STRUCTURE

PROGRAMMING SOLVING KNOWLEDGE



Typical job duties for data scientists

There's not a definitive job description when it comes to a data scientist role. But here are a few things you'll likely be doing:

- Collecting large amounts of unruly data and transforming it into a more usable format.
- Solving business-related problems using data-driven techniques.
- Working with a variety of programming languages, including SAS, R and Python.
- Having a solid grasp of statistics, including statistical tests and distributions.
- Staying on top of analytical techniques such as machine learning, deep learning and text analytics.
- Communicating and collaborating with both IT and business.
- Looking for order and patterns in data, as well as spotting trends that can help a business's bottom line.

Why the Hype Around Data Science?

- The demand for data scientists will soar by 28% by 2023
- Data scientist roles have grown over 650% since 2012, but currently, 35,000 people in the US have data science skills, while hundreds of companies are hiring for those roles.
- Software engineering is a common starting point for professionals who are in the top five fastest growing jobs today.
- Data Science gives you career flexibility

What is Machine Learning?

Learn from experience



Learn from experience



Follow instructions



What is Machine Learning ?

Machine learning teaches computers to do what comes naturally to humans and animals: learn from experience. Machine learning algorithms use computational methods to “learn” information directly from data without relying on a predetermined equation as a model. The algorithms adaptively improve their performance as the number of samples available for learning increases.

A Definition

A computer program is said to learn from experience E with respect to some task T and some performance measure P if its performance on T , as measured by P , improves with experience E .

-Tom Mitchell

A Small Question

Suppose we feed a learning algorithm a lot of historical weather data, and have it learn to predict weather. In this setting, what is T, P, E ?

ARTIFICIAL INTELLIGENCE

Early artificial intelligence stirs excitement.



MACHINE LEARNING

Machine learning begins to flourish.



DEEP LEARNING

Deep learning breakthroughs drive AI boom.



1950's

1960's

1970's

1980's

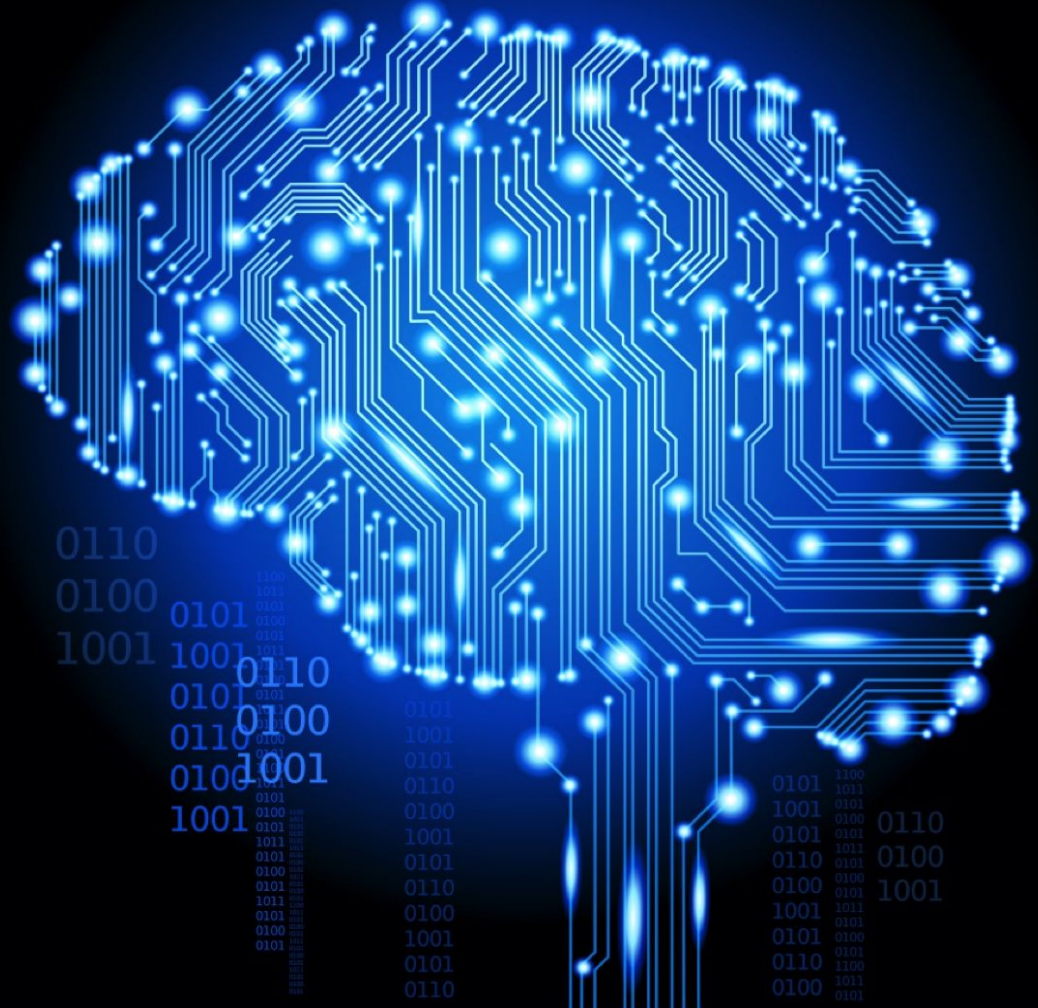
1990's

2000's

2010's

Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

More Data,
More Questions,
Better Answers



Real World Applications

With the rise in big data, machine learning has become particularly important for solving problems in areas like these:

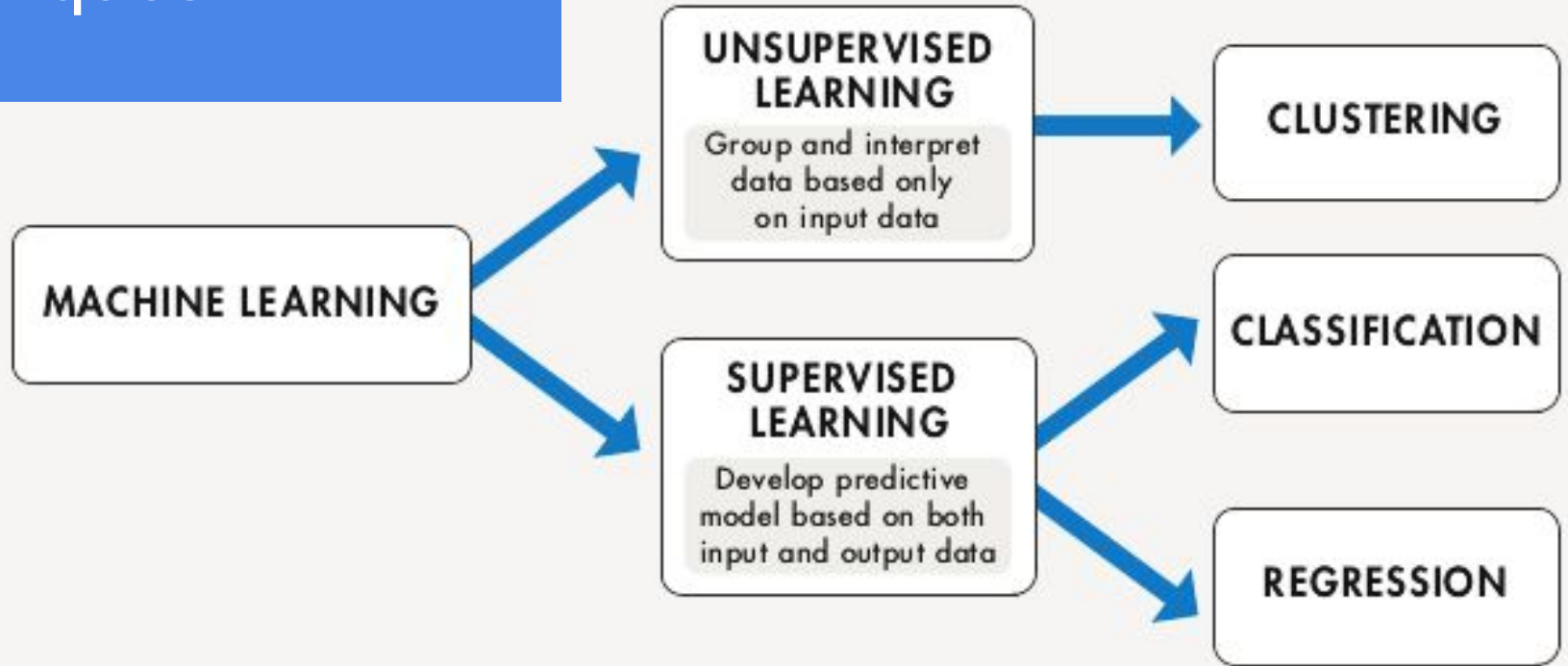
- Image processing and computer vision, for face recognition, motion detection, and object detection
- Computational biology, for tumor detection, drug discovery, and DNA sequencing
- Energy production, for price and load forecasting
- Automotive, aerospace, and manufacturing, for predictive maintenance
- Natural language processing

How Machine Learning Works

Machine learning uses two types of techniques:

- Supervised learning, which trains a model on known input and output data so that it can predict future outputs
- Unsupervised learning, which finds hidden patterns or intrinsic structures in input data.

Machine Learning Techniques



Supervised Learning

The aim of supervised machine learning is to build a model that makes predictions based on evidence in the presence of uncertainty. A supervised learning algorithm takes a known set of input data and known responses to the data (output) and trains a model to generate reasonable predictions for the response to new data

Classification - predict discrete responses

Classification models classify input data into categories. for example, whether an email is genuine or spam, or whether a tumor is cancerous or benign.

Regression - predict continuous responses

for example, changes in temperature or fluctuations in power demand. Typical applications include electricity load forecasting and algorithmic trading.

Unsupervised Learning

Unsupervised learning finds hidden patterns or intrinsic structures in data. It is used to draw inferences from dataset consisting of input data without labeled responses.

Clustering is the most common unsupervised learning technique. It is used for exploratory data analysis to find hidden patterns or groupings in data. Applications for clustering include gene sequence analysis, market research, and object recognition

Knowledge Test

Which of the following would you apply supervised learning to?

1. Given genetic (DNA) data from a person, predict the odds of him/her developing diabetes over the next 10 years.
2. Given a large dataset of medical records from patients suffering from heart disease, try to learn whether there might be different clusters of such patients for which we might tailor separate treatments.
3. Given data on how 1000 medical patients respond to an experimental drug (such as effectiveness of the treatment, side effects, etc.), discover whether there are different categories or "types" of patients in terms of how they respond to the drug, and if so what these categories are.
4. Have a computer examine an audio clip of a piece of music, and classify whether or not there are vocals (i.e., a human voice singing) in that audio clip, or if it is a clip of only musical instruments (and no vocals).

Knowledge Test

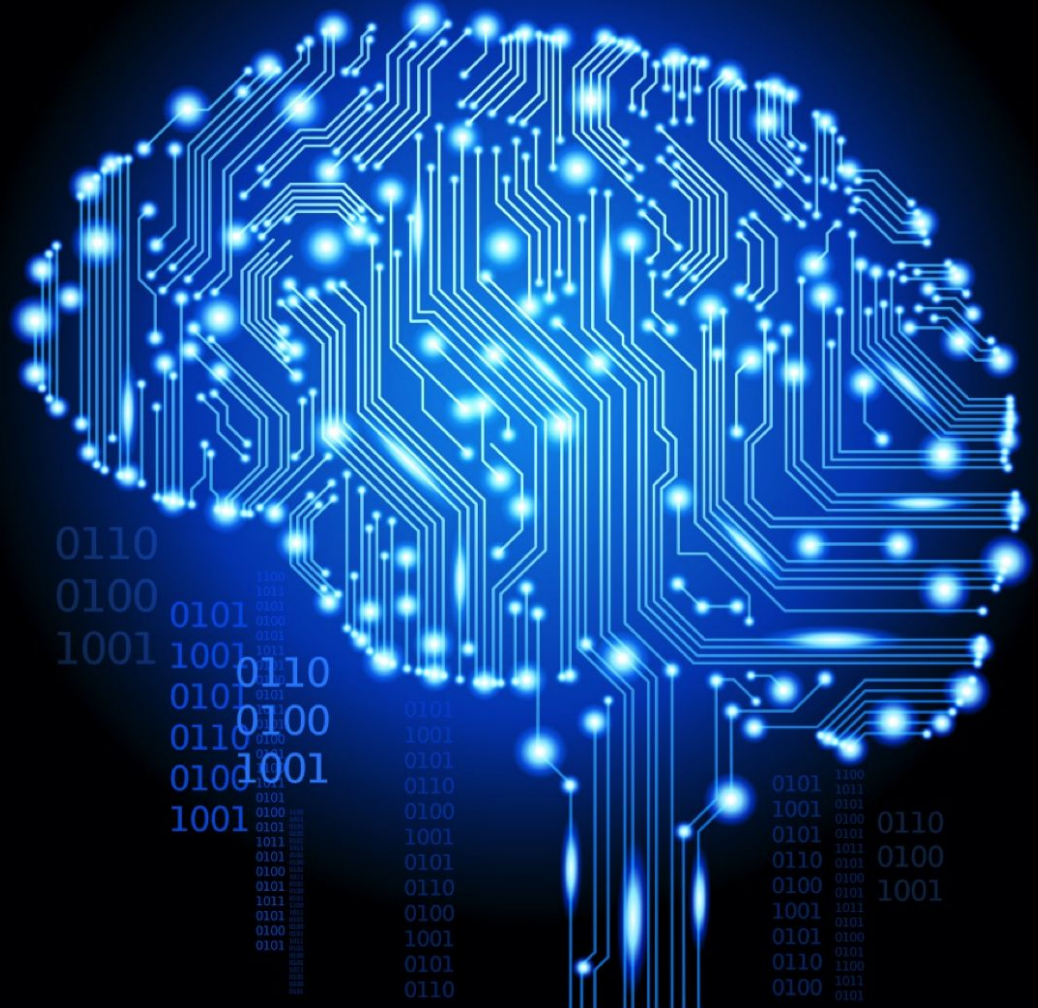
Which of the following questions can be answered using a classification algorithm?

1. How does the exchange rate depend on the GDP?
2. Does a document contain the handwritten letter S?
3. How can I group supermarket products using purchase frequency?

Knowledge Test

1. Suppose you are working on weather prediction, and you would like to predict whether or not it will be raining at 5pm tomorrow. You want to use a learning algorithm for this. Would you treat this as a classification or a regression problem?
2. Suppose you are working on stock market prediction. You would like to predict whether or not a certain company will declare bankruptcy within the next 7 days (by training on data of similar companies that had previously been at risk of bankruptcy). Would you treat this as a classification or a regression problem?

How Do You
Decide Which
Algorithm
to Use?



Choosing the right algorithm can seem overwhelming

There are dozens of supervised and unsupervised machine learning algorithms, and each takes a different approach to learning.

The background is a dark blue gradient with a complex network of thin, light blue lines connecting small, semi-transparent blue dots. The dots are scattered across the frame, with a higher density of lines and dots towards the bottom, creating a sense of depth and connectivity.

There is no best method or one size fits all. Finding the right algorithm is partly just trial and error

But algorithm selection also depends on the size and type of data you're working with, the insights you want to get from the data, and how those insights will be used.

Two - Class Classification



Two-class SVM

Under 100 features,
linear model



Two-class averaged perceptron

Fast training, linear
model



Two-class Bayes point machine

Fast training, linear
model



Two-class decision jungle

Accurate, small
memory footprint



Two-class locally deep SVM

Under 100
features



Two-class decision forest

Accurate, fast
training



Two-class logistic regression

Fast training, linear
model



Two-class boosted decision tree

Accurate, fast training,
large memory footprint



Two-class neural network

Accurate, long
training times

Multi - Class Classification



Multiclass logistic regression

Fast training times,
linear model



Multiclass neural network

Accuracy, long
training times



Multiclass decision forest

Accuracy, fast
training times



Multiclass decision jungle

Accuracy, small
memory footprint



One-v-all multiclass

Depends on the
two-class classifier

Anomaly Detection

One Class SVM

Under 100 features, aggressive boundary



PCA-Based Anomaly Detection

Fast training times



Regression



Ordinal Regression

Data in rank ordered categories



Poisson Regression

Predicts event counts



Fast forest quantile regression

Predicts a distribution



Linear Regression

Fast training, linear model



Bayesian Linear Regression

Linear model, small data sets



Neural Network Regression

Accurate, long training times



Decision Forest Regression

Accurate, fast training times



Boosted Decision Tree Regression

Accurate, fast training times, large memory footprint

Clustering



K-Means

Unsupervised learning

When should we use Machine Learning

Consider using machine learning when you have a complex task or problem involving a large amount of data and lots of variables, but no existing formula or equation.

Hand-written rules and equations are too complex—as in face recognition and speech recognition.



The rules of a task are constantly changing—as in fraud detection from transaction records.



The nature of the data keeps changing, and the program needs to adapt—as in automated trading, energy demand forecasting, and predicting shopping trends.



Knowledge Test

Have a look at the statements below and identify the one which is not a machine learning problem

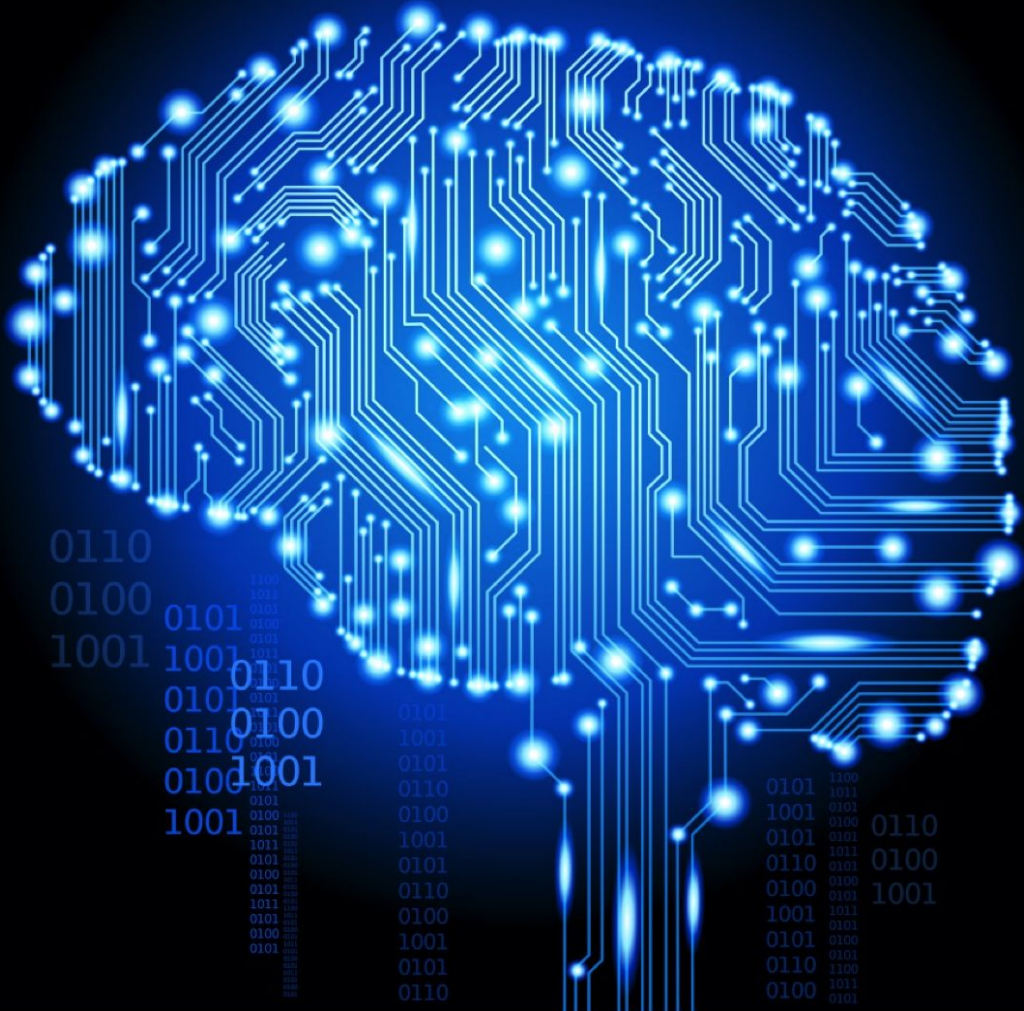
1. Given a viewer's shopping habits, recommend a product to purchase the next time she visits your website.
2. Given the symptoms of a patient, identify her illness.
3. Predict the USD/EUR exchange rate for February 2023.
4. Compute the mean wage of 10 employees for your company.

Knowledge Test

Which of the following statements uses a machine learning model?

1. Determine whether an incoming email is spam or not
2. Obtain the name of last year's FIFA Ballon d'Or champion
3. Automatically tagging your new Facebook photos
4. Select the student with the highest grade on a statistics course

Getting Started



There is NO Straight Line

With machine learning there's rarely a straight line from start to finish. You'll find yourself constantly iterating and trying different ideas and approaches

Machine Learning Challenges

- Data comes in all shapes and sizes
- Preprocessing your data might require specialized knowledge and tools
- It takes time to find the best model to fit the data.

Questions to Ask Before Starting

Every machine learning workflow begins with three questions:

- What kind of data are you working with?
- What insights do you want to get from it?
- How and where will those insights be applied?

Your answers to these questions help you decide whether to use supervised or unsupervised learning.

Data Science - Five Questions

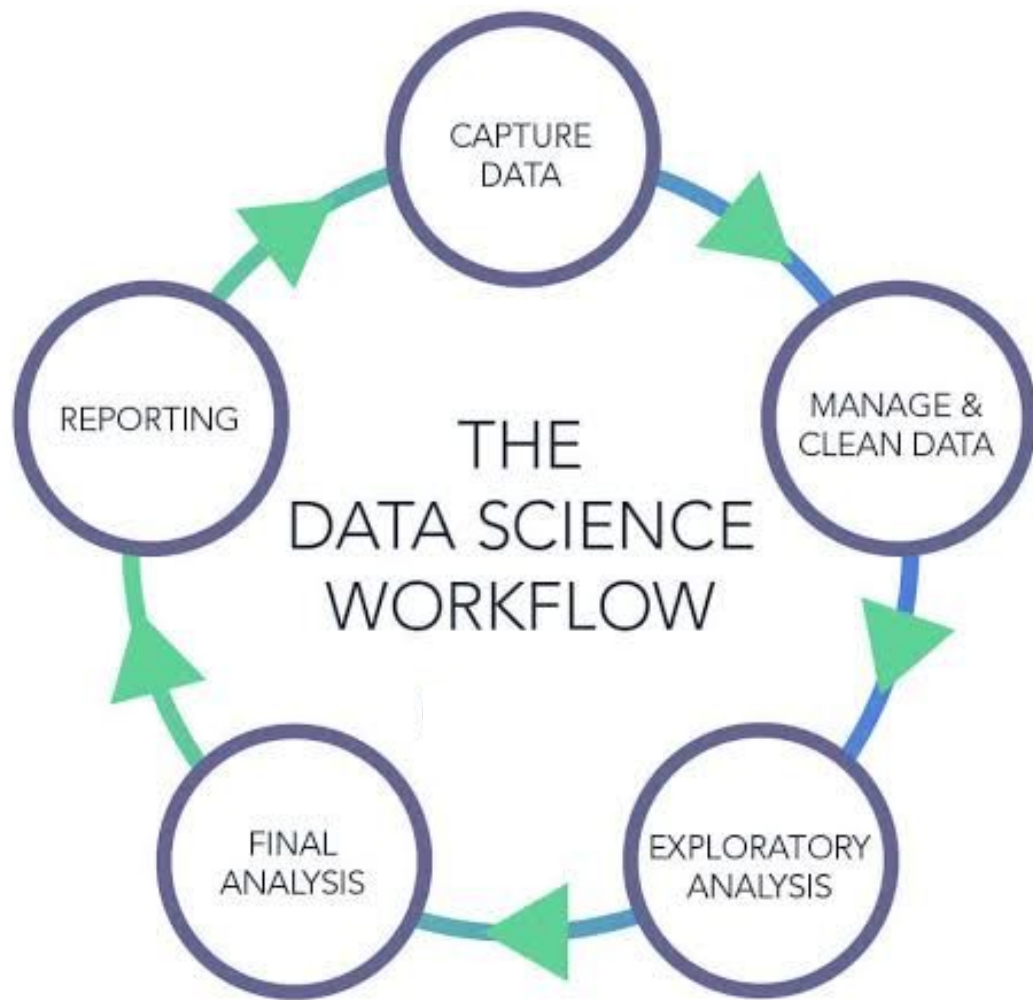
There are only five questions that data science answers:

- Is this A or B?
- Is this weird?
- How much – or – How many?
- How is this organized?
- What should I do next?

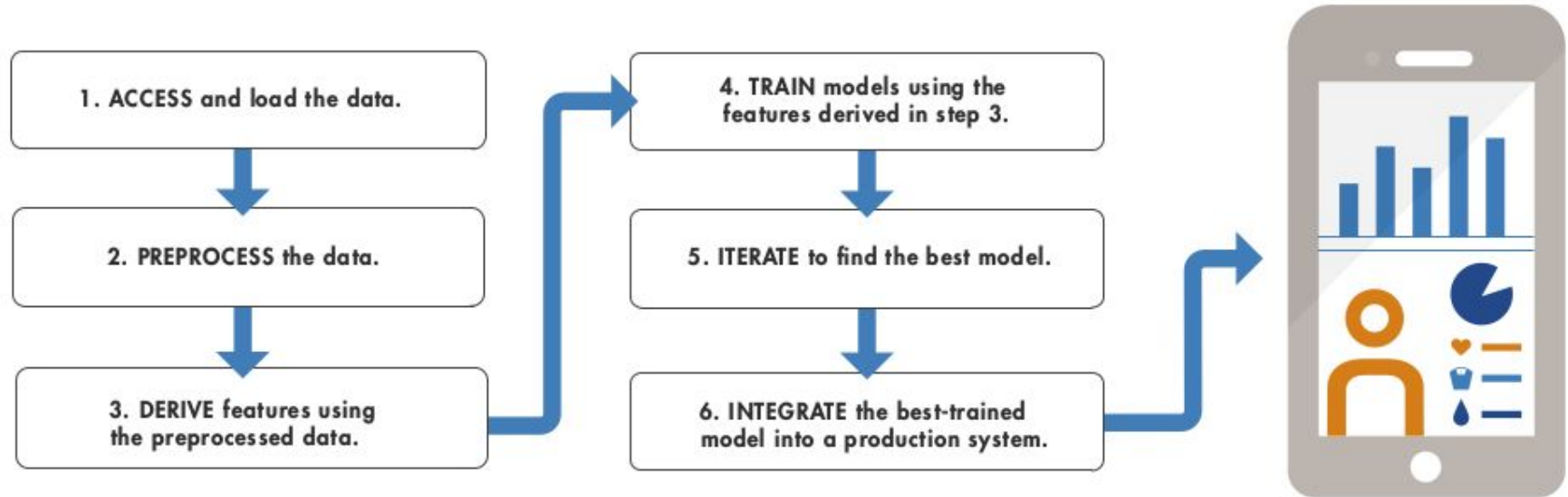
Knowledge Test

Which of the following questions can be answered using a classification algorithm?

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Workflow at a Glance



Step 1 - Load the Data

We store the labeled data sets in a text file. A flat file format such as text or CSV is easy to work with and makes it straightforward to import data.

Machine learning algorithms aren't smart enough to tell the difference between noise and valuable information. Before using the data for training, we need to make sure it's clean and complete

Step 2 - Preprocess the Data

To preprocess the data we do the following:

- Look for outliers—data points that lie outside the rest of the data
- Check for missing values
- Divide the data into two sets
 - We save part of the data for testing (the test set) and use the rest (the training set) to build models. This is referred to as holdout, and is a useful cross-validation technique

Step 3 - Derive Features

Deriving features (also known as feature engineering or feature extraction) turns raw data into information that a machine learning algorithm can use.

Use feature selection to:

- Improve the accuracy of a machine learning algorithm
- Boost model performance for high-dimensional data sets
- Improve model interpretability
- Prevent overfitting

Step 4 - Build and Train Model

- The predefined algorithms and the test data are used for building the model.
- The training data is used to train and evaluate the model

Step 5 - Improve the Model

Improving a model can take two different directions: make the model simpler or add complexity.

Simplify - reduce the number of features

Add Complexity - make it more fine-tuned

Simplify

Popular feature reduction techniques include:

- **Correlation matrix** – shows the relationship between variables, so that variables (or features) that are not highly correlated can be removed.
- **Principal component analysis (PCA)** - eliminates redundancy by finding a combination of features that captures key distinctions between the original features and brings out strong patterns in the dataset.
- **Sequential feature reduction** – reduces features iteratively on the model until there is no improvement in performance

Add Complexity

- **Use model combination** – merge multiple simpler models into a larger model that is better able to represent the trends in the data than any of the simpler models could on their own.
- **Add more data sources**